Deliverable: 1 page called "Triage Rules."

Task 2 (PARALLEL): Write the mapping rubric
How you assign aligned/silent/conflict without "AI magic."

- *Use the guideline topics from Team B*
- *Map each evidence cluster to a topic*
- *Apply rubric:*
    - *aligned if evidence supports what guideline recommends*
      conflict if evidence pushes against a guideline "do not"
    - *silent if guideline does not address it or is too vague*
      Deliverable: 1 page called "Mapping Rubric + Examples."

TEAM C CONTRIBUTION BELOW ⬇️

Team C Deliverable: Data Modeling aligned to Team B data (Deterministic ERS + Topic Mapping)

Purpose

Rank NSCLC evidence items deterministically for a given patient vignette and map each ranked item to Team B's **Guideline Topic Catalog**, producing:

1. a ranked "Top Evidence" list (triage), and
2. a mapping label per item: **aligned / guideline-silent / conflict** (mapping).

This MVP does not recommend treatment. It ranks and labels evidence relative to guideline topics.

Inputs consumed from Team B

B1. Patient Vignette Schema (what the user provides)

Team C assumes every vignette contains these fields (no "unknown" in MVP vignettes unless explicitly in the stress set):

- cancer_type: NSCLC (fixed)
- disease_setting: early | locally_advanced | metastatic
- histology: adenocarcinoma | squamous
- performance_status: 0 | 1 | 2 | 3 | 4
- biomarkers:
    - EGFR: yes | no

    - ALK: yes | no
    - ROS1: yes | no
    - PD_L1_bucket: lt1 | 1to49 | ge50

These vignette fields are used for the **Clinical Relevance Gate**.

B2. Evidence Record Schema (what Team B extracts from PubMed, plus normalization)

Team C expects Team B to provide each evidence item with:

- evidence_id (stable)
- title
- publication_year
- evidence_type: guideline | systematic_review | phase3_rct | phase2_rct | prospective_obs | retrospective | case_series | expert_opinion
- relevant_N (integer, or null if not available)
- source_category: guideline_body | high_impact_journal | specialty_journal | preprint | industry_whitepaper
- population_tags:
    - disease: NSCLC
    - disease_setting: early | locally_advanced | metastatic | mixed
    - histology: adenocarcinoma | squamous | mixed | all_nsclc
    - biomarkers: EGFR/ALK/ROS1/PDL1 applicability tags when available (otherwise “unspecified”)
- intervention_tags (free text tags or controlled list)
- outcome_tags (OS | PFS | response | toxicity | QoL etc., if available)

If a field is missing in source data, Team B may fill it via simple deterministic extraction rules, or leave null and apply scoring defaults (described below).

B3. Guideline Topic Catalog (the anchor for mapping)

Team C assumes Team B produces a small catalog of 10–15 topics with a stable identifier:

Each topic must contain:

- topic_id
- topic_title (human readable)
- topic_applicability:
    - disease_setting (one or more)
    - histology (one or more)
    - biomarker_conditions (simple yes/no rules; example: “EGFR=no AND ALK=no AND ROS1=no AND PD-L1=ge50”)
- topic_intervention_tags (keywords or controlled tags that connect evidence to this topic)
- guideline_stance (from curated excerpt set):
    - recommend | conditional | do_not_recommend | not_covered

- stance_notes (short text; optional)
- prerequisites (optional list; can be empty for MVP)

This catalog is Plan A if you have real guideline excerpts; Plan B is a curated excerpt/synthetic topic set created by the team.

Model Overview (deterministic steps)

Step 1. Clinical Relevance Gate (hard gate)

An evidence item is eligible for ranking only if it matches the vignette on:

1. disease match: evidence.population_tags.disease includes NSCLC
2. setting match: evidence.population_tags.setting matches vignette.disease_setting OR is “mixed” and still applicable
3. histology match: evidence.population_tags.histology is vignette.histology OR “all_nsclc” OR “mixed”
4. biomarker compatibility:

- if evidence is biomarker-specific (example: EGFR-mutant only), then vignette must satisfy that biomarker condition
- if evidence biomarker applicability is “unspecified”, it passes gate but receives no extra match benefit later

If an item fails the gate, it is not ranked. It may be shown under “Secondary references (not applicable)” with the reason.

Step 2. Evidence Ranking Score (ERS-MVP, 0–60)

For each eligible evidence item, compute:

A) Evidence strength & methodology (0–20)

- guideline: 20
- systematic_review: 18
- phase3_rct: 16
- phase2_rct: 13
- prospective_obs: 10
- retrospective: 6
- case_series / expert_opinion: 2

B) Dataset robustness (0–15) using relevant_N

- N ≥ 300: 15
- 100–299: 12
- 50–99: 9
- 20–49: 6

- < 20: 3
- if relevant_N is null: default 6 (mid, conservative)

C) Source credibility (0–15)

- guideline_body: 15
- high_impact_journal: 12
- specialty_journal: 9
- preprint: 5
- industry_whitepaper: 2
- if source_category is null: default 6

D) Recency (0–10) using publication_year (relative to current year)

- ≤3 years: 10
- 4–6 years: 6
- 7–10 years: 3
- 10 years: 0

ERS-MVP = A + B + C + D

Sorting: descending ERS-MVP, with deterministic tie-breakers:

1. higher A wins
2. newer year wins
3. larger N wins
4. otherwise keep both and cluster by topic

Default inclusion threshold:

- include in “Top Evidence” if ERS-MVP ≥ 30
- show below threshold as “Secondary reference”

Step 3. Topic Mapping (evidence → guideline topic)

For each ranked evidence item:

1. find candidate topics where:
    - vignette satisfies topic_applicability rules, and
    - evidence.intervention_tags overlaps topic_intervention_tags
2. choose the best topic match:
    - highest tag overlap count wins
    - tie-breaker: topic with most specific biomarker conditions wins
3. if no topic matches: mapping label is “guideline-silent” and topic is “none”

Step 4. Mapping Label (aligned / guideline-silent / conflict)

Using the chosen topic’s guideline_stance:

- aligned:

    - topic stance is recommend or conditional, and evidence supports that intervention category in an applicable population
- conflict:
    - topic stance is do_not_recommend, but evidence suggests benefit for that same intervention in applicable population
- guideline-silent:
    - topic stance is not_covered, OR no topic match was found

For MVP, “supports” is implemented deterministically as:

- evidence_type guideline/systematic_review/phase3_rct is considered “supporting” if it matches topic intervention_tags and passes gate
- you can refine later with directionality extraction, but MVP stays conservative

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MVP Output Format (what appears in UI / report)

For each evidence item in the top list:

- rank position
- evidence_id, title, year
- ERS-MVP total + breakdown (A/B/C/D)
- mapped topic_id + topic_title (or none)
- mapping label: aligned / guideline-silent / conflict
- short applicability note (generated from field comparisons, not free-text generation)

Worked example aligned to Team B vignette fields and topic catalog

Example vignette (NSCLC MVP)

- disease_setting: metastatic
- histology: adenocarcinoma
- PS: 1
- EGFR: no, ALK: no, ROS1: no
- PD-L1: ge50

Example topic from Team B catalog (illustrative)

- topic_id: TB_TOPIC_07
- topic_title: “Metastatic NSCLC, driver-negative, PD-L1 ≥50: first systemic therapy evidence”
- topic_applicability: metastatic + adenocarcinoma + EGFR=no AND ALK=no AND ROS1=no AND PD-L1=ge50

- topic_intervention_tags: “PD-1”, “PD-L1”, “immunotherapy”, “pembrolizumab” (example tags)
- guideline_stance: recommend

Now any evidence item that:

- passes gate for this vignette, and
- has intervention_tags overlapping those tags,
  will map to TB_TOPIC_07 and likely be labeled aligned (unless stance was do_not_recommend).